

Taste bud

The **gustatory system** or **sense of taste** is the [sensory system](#) that is partially responsible for the [perception](#) of taste.^[1] Taste is the perception stimulated when a substance in the [mouth reacts chemically](#) with [taste receptor](#) cells located on [taste buds](#) in the [oral cavity](#), mostly on the [tongue](#). Taste, along with [the sense of smell](#) and [trigeminal nerve](#) stimulation (registering texture, pain, and temperature), determines [flavors](#) of [food](#) and other substances. [Humans](#) have taste receptors on taste buds and other areas, including the upper surface of the [tongue](#) and the [epiglottis](#).^{[2][3]} The [gustatory cortex](#) is responsible for the perception of taste.

The tongue is covered with thousands of small bumps called [papillae](#), which are visible to the [naked eye](#).^[2] Within each papilla are hundreds of taste buds.^{[1][4]} The exceptions to this is the [filiform papillae](#) that do not contain taste buds. There are between 2000 and 5000^[5] taste buds that are located on the back and front of the tongue. Others are located on the roof, sides and back of the mouth, and in the [throat](#). Each taste bud contains 50 to 100 taste receptor cells.

Taste receptors in the mouth sense the five basic tastes: [sweetness](#), [sourness](#), [saltiness](#), [bitterness](#), and [savoriness](#) (also known as *savory* or *umami*).^{[1][2][6][7]} Scientific experiments have demonstrated that these five tastes exist and are distinct from one another. Taste buds are able to tell different tastes apart when they interact with different molecules or ions. Sweetness, savoriness, and bitter tastes are triggered by the binding of molecules to [G protein-coupled receptors](#) on the [cell membranes](#) of taste buds. Saltiness and sourness are perceived when [alkali metals](#) or [hydrogen ions](#) meet taste buds, respectively.^{[8][9]}

The basic tastes contribute only partially to the sensation and flavor of food in the mouth—other factors include [smell](#),^[1] detected by the [olfactory epithelium](#) of the nose;^[10] [texture](#),^[11] detected through a variety of [mechanoreceptors](#), muscle nerves, etc.;^[12] temperature, detected by [temperature receptors](#); and "coolness" (such as of [menthol](#)) and "hotness" ([pungency](#)), by [chemesthesis](#).

